

Markov Chain.

Discrete-time Markov Chain.

An discrete-time Markov Chain is an integer sequence of Random Variable, which only correlated to its previous step (Markov properties), which allows simplified calculation of long time transition probability.

Def. A sequence of R.V. $(X_n)_{n \geq 0}$, where $X_n: \Omega \rightarrow S \subset \mathbb{Z}$ is called a Markov Chain with initial distribution λ if

~~and~~ a) X_0 has distribution λ

b) satisfied Markov property.

$$P(X_{n+1} = i_{n+1} | X_n = i_n, \dots, X_0 = i_0) = P(X_{n+1} = i_{n+1} | X_n = i_n).$$

Remark: X_n can denote the state of the chain at a certain time n .

- S are referred as state space, which represent all possible outcome of each state.
- If denote $P(X_{n+1} = i_{n+1} | X_n = i_n) = P_{i_n i_{n+1}}$, then for finite S , $P \in \mathbb{R}^{|S| \times |S|}$ are referred as transition matrix, and $P_{i_n i_{n+1}}$ represent the transition probability from state i_n to i_{n+1} .

prob: $P_{ij} > 0, \sum_{j \in S} P_{ij} = 1$

Next we looky into n -step transition probability.
 The Markov property and total probability formula allows us to calculate n step transition probs using single step transition matrix.

Probability of n -step path.

we wish to calculate the probability that the n -step state is at j while start from i , or further the probability the state is at j on n -step with any initial state.

The calculation is straight forward using total probability and Markov property.

Th. ~~let~~ $(X_n)_{n \geq 0}$ be a Markov chain with initial distribution λ and transition matrix P . Then for all $n, m \geq 0$.

$$1), P(X_n = j | X_0 = i) = (P^n)_{ij}$$

$$2) P(X_n = i) = (\lambda P^n)_i$$

Proof: ~~$P(X_n = j | X_0 = i)$~~

$$\begin{aligned} P(X_n = j | X_0 = i) &= \sum_{i_0 \in S} \cdots \sum_{i_{n-1} \in S} P(X_n = j | X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \\ &\quad \cdot P(X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \\ &= \sum_{i_0 \in S} \cdots \sum_{i_{n-1} \in S} P(X_n = j | X_{n-1} = i_{n-1}) \cdot P(X_{n-1} = i_{n-1} | X_{n-2} = i_{n-2}, \dots, X_0 = i_0) \\ &\quad \cdot P(X_{n-2} = i_{n-2}, \dots, X_0 = i_0) \\ &= \sum_{i_0 \in S} \cdots \sum_{i_{n-1} \in S} P(X_n = j | X_{n-1} = i_{n-1}) \cdot P(X_{n-1} = i_{n-1} | X_{n-2} = i_{n-2}) \cdots \\ &\quad \cdots P(X_1 = i_1 | X_0 = i_0) \cdot P(X_0 = i_0) \\ &= \sum_{i_0 \in S} \cdots \sum_{i_{n-1} \in S} \lambda_{i_0} P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} j} = (\lambda P^n)_j. \end{aligned}$$

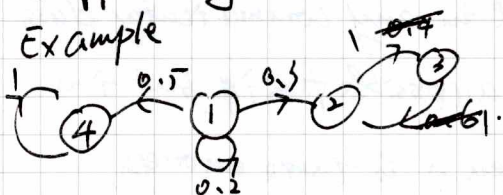
$$\begin{aligned}
 P(X_n=j | X_0=i) &= \sum_{x_{n-1}=s} \dots \sum_{x_1=t_1} P(X_n=j | X_{n-1}=t_{n-1}, \dots, X_1=t_1, X_0=i) \\
 &= \sum_{x_{n-1}=s} \dots \sum_{x_1=t_1} P(X_n=j | X_{n-1}=t_{n-1}) \cdot P(X_{n-1}=t_{n-1} | X_{n-2}=t_{n-2}) \\
 &\quad \dots P(X_1=t_1 | X_0=i) \\
 &= \sum_{x_{n-1}=s} \dots \sum_{x_1=t_1} P_{j, t_{n-1}} P_{t_{n-1}, t_{n-2}} \dots P_{t_1, i} \\
 &= (p^n)_j
 \end{aligned}$$

Remark: using matrix has assume finite state space.
the summation however still true when state is infinite

(2) can be extend to $P(X_{nm}=j | X_n=i) = (p^m)_j$

We then look into the asymptotic behavior of the state probability distribution, i.e. when $n \rightarrow \infty$, will the transition between state tends to equilibrate. Such behavior of Markov Chain will be the foundation behind Monte Carlo based method.

Markov chains different state in Markov Chain behave differently when $n \rightarrow \infty$. The following give an simple example.



Noticed the following:

①. I can access 2, 3, & 4, while 2, 3, 4 can not access 1.

②. Once state enter state 2, it will only transition between {2, 3}.
similar for 4

③ when n is large enough, the state will be "trapped" to {4, 3} or {2, 3}.

We can have the following definition to formalise this observation.
Def: we say that i and j communicate. ~~if $\exists$ finite sequence~~
sequence $i \dots i$ s.t. $\iff i$ lead to j and j lead i .

Def we say state i lead to j (or j is accessible from i)
if $\exists$ finite sequence $i, i_1, \dots, i_n$ s.t.

$$P_{i, i_1} > 0, P_{i_1, i_2} > 0, \dots, P_{i_{n-1}, i_n} > 0 \iff P(X_n = j, \tau_0^{j,0} | X_0 = i) > 0.$$

Def a set of states S is called communicating classes
or recurrent class if $\forall i, j \in S, i$ communicate
with j . (or strongly connected component if
view Markov chain as graph).

Def A communicating class C is closed if.
 $\forall i \in C, \text{ if } i \text{ lead to } j, \text{ then } j \in C$

Remark: This means once the state enter the closed
class, it will not come out.

A absorbing state is simply τ as
a closed class.

Noticed that if a state i is not in any communicating class,
that means $\exists$ state j s.t. j is accessible from i but i is
not accessible from j . This means when n is large enough,
 $i \rightarrow j$ transition will happen and after that the state will
not come back. Therefore, next we will focus on MC
with single communicating class. (irreducible).

Stationary / invariant distribution

we first discuss the problem: for a ~~linear~~ irreducible MC Q , will the probability of being in each state tends to be approximately 'stable' in the long run?

First we define what ~~we~~ we mean by 'stable'. Essentially, hope that the prob dist of ~~each~~ state won't change after transition. Thus come the following definition.

Def. Let λ denote the probability distribution of state X .

i.e. $\lambda_i = P(X=i)$.

we say λ is a stationary dist if.

$$\lambda = \lambda P, \quad \sum \lambda_i = 1$$

Remark: the definition means the ^{1 step} transition of current probability $\sum_j \lambda_j P_{ji} = \lambda_i$ remain the same.

It can be shown that if the transition probability converge ^{to stable} to some number invariant to ~~previous state~~, then initial state, then the converged distribution is stable / invariant. (for finite state space).

Th. let the set of state S of MC $\{X_n\}_{n \geq 0}$ be finite, and for some $i \in S$, $P(X_n = j | X_0 = i) = (P^n)_{ij} \rightarrow \pi_j$ as $n \rightarrow \infty$ $\forall j \in S$. then π is an invariant distribution.

proof: $\pi_j = \lim_{n \rightarrow \infty} (P^n)_{ij} = \lim_{n \rightarrow \infty} \sum_{k \in S} (P^{n-1})_{ik} P_{kj} = \sum_{k \in S} \lim_{n \rightarrow \infty} (P^{n-1})_{ik} P_{kj} \stackrel{?}{=} \sum_{k \in S} \pi_k P_{kj}$

$$\sum_{j \in S} \pi_j = \sum_{j \in S} \lim_{n \rightarrow \infty} (P^n)_{ij} = \lim_{n \rightarrow \infty} \sum_{j \in S} (P^n)_{ij} = 1.$$

thus $\pi = \lim_{n \rightarrow \infty} (P^n)_{ij}$ is stationary distribution

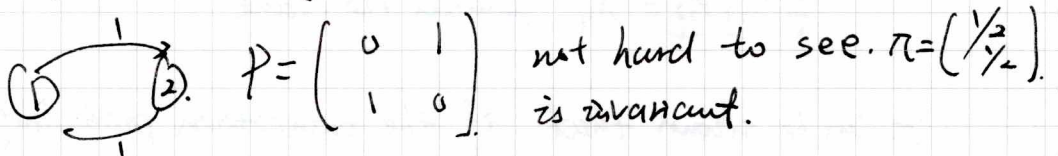
Remark: • the interchange of limit and summation is permitted for finite space.

• the ~~TH~~ ~~invariant~~, guarantee the converged dist is invariant, the other way around might not be true.

• ~~The convergent probability to have might be depending on the initial state.~~

• for the probability to converge, ~~the~~ need to restrict to single communicating class, otherwise the $\lim_{n \rightarrow \infty} (P^n)_{ij}$ will be depending on the initial state as they might lead to different recurrent class.

Example: this simple example show that even for a single communicating class, existence of invariant dist does not guarantee convergence.



however. the probability oscillate and never converge.

$$P^{2k} = I, \quad P^{2k+1} = P.$$

~~the~~ We then further exclude such case. ~~in the strategy~~ by define the aperiodic property of MC.

Def state i is called aperiodic if $(P^{(n)})_{ii} > 0$ for all sufficiently large n .

Def A Markov Chain is called aperiodic. if $\forall i \in S$ is aperiodic.

Th. Suppose a MC with set of state is irreducible and has a n aperiodic state i . Then for all state $j, k \in S$, $(P^n)_{jk} > 0$ for all sufficient large n . In particular, all states are aperiodic.

proof: Since MC is irreducible, $\exists r, s$ s.t. $(P^r)_{ji} > 0$ and $(P^{os})_{ik} > 0$. Then for sufficiently large n .

$$\begin{aligned} (P)^{n+r+s}_{jk} &= \sum_{i_1, \dots, i_n \in S} (P)^r_{ji_1} P_{i_1 i_2} P_{i_2 i_3} \dots P_{i_n i_n} (P)^s_{i_n k} \\ &= \sum_{i_1, i_n \in S} (P)^r_{ji_1} (P)^n_{i_1 i_n} (P)^s_{i_n k} \\ &\geq \sum_{i_1, i_n \in S} (P)^r_{ji_1} (P^n)_{i_1 i_n} (P)^s_{i_n k} > 0. \end{aligned}$$

For an aperiodic MC, the existence of invariant dist will guarantee convergence to it.

Th. Suppose that $(X_n)_{n \geq 0}$ is an irreducible and aperiodic MC, with transition matrix P and initial dist π .

Suppose P has an invariant dist π , Then.

$$P(X_n = j) \rightarrow \pi_j \text{ as } n \rightarrow \infty \quad \forall j \in S.$$

In particular, $(P^n)_{ij} \rightarrow \pi_j$ as $n \rightarrow \infty \quad \forall i, j \in S$.

Further more, for finite state space, we had the following

Th. Suppose $(X_n)_{n \geq 0}$ is an irreducible and aperiodic MC with finite set of state. Then $\exists$ unique invariant dist π and $P(X_n = j) \rightarrow \pi_j$ for any initial dist π .